

6.1 Model versioning

Logistic Regression Model for predicting accidents and a Linear Regression model for guessing traffic volume. These models are very basic and run quickly, but they make a lot of mistakes because they only look at data in a straight line.

Random Forest Models. These models are much smarter because they combine many decision trees together.

6.2 Experiment Tracking

Parameters: It remembers the exact settings we chose for our models, like how deep our decision trees was or how many trees we used.

Metrics: It automatically logs the final scoreboard math, so we can see which run performed best.

Model Versions: Every time we retrain the code with fresh data, MLflow saves the new code file asset separately so we never lose our old work.

Experiments: It groups all our test runs under a single project folder so we can compare our scores on a clean history chart.

6.3 Deployment

- Instead of running the code the script runs in the background as a live web link.
- An external user or a traffic mobile app sends a simple text packet to our link containing the current travel details.
- Our model instantly reads these values, runs them through our trained Random Forest formula, and sends back a clean text prediction message within milliseconds.

6.4 Monitoring

- We constantly compare our model guess against the actual number of cars count on the highway. If the number starts growing bigger over time, it means our AI is getting rusty and losing its touch.
- We check if the incoming real-world data changes completely compared to the data we used to train the model.

6.5 Alerting

-This status lights up in green when the model prediction error stays low and healthy. It means the AI is guess traffic correctly and everything is running perfectly.

-This status alert in red when our prediction error cross a dangerous threshold limit. It instantly alerts the engineering team.